

Supplementary Information

Cost–Resilience Thresholds for Battery–Hydrogen Hybrid Microgrids in Islanded Power Systems

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Supplementary file caption

This supplementary file documents the fixed-portfolio definitions, synthetic-load construction, outage-setting rationale, screened parameter ranges, and reproducibility workflow used in the main manuscript.

S1. Fixed portfolio definitions and design intent

Table S1 reports the fixed capacities used in the three candidate portfolios. These values are taken directly from the scenario configuration used to generate the submitted results.

Table 1: Fixed capacities used in the candidate portfolios.

Portfolio	Label	PV (MW)	Battery power (MW)	Battery energy (MWh)	Diesel (MW)	E
battery_only	PV + battery	7.0	2.2	24.0	0.0	
diesel_battery	PV + diesel + battery	2.8	1.5	8.0	2.4	
battery_hydrogen	PV + battery + hydrogen	14.0	2.0	12.0	0.0	

The battery–hydrogen portfolio additionally includes a hydrogen tank with a fixed inventory capacity of 50,000 kg. The fixed-capacity design is intentional: it allows the manuscript to compare operational and resilience behavior across candidate portfolios under common assumptions without claiming endogenous optimal sizing.

S2. Synthetic load archetype and case-study inputs

The case-study inputs were generated from the scenario file and the prepared 8760-hour dataset used by the dispatch model. The synthetic load archetype is calibrated using the following inputs:

- peak demand = 1.65 MW
- load factor = 0.72
- evening peak strength = 0.30
- morning peak strength = 0.10

- commercial load fraction = 0.14
- tourism multiplier = 1.08 in March, April, and December
- seasonal monthly multipliers = [1.02, 1.03, 1.04, 1.05, 1.04, 1.02, 0.99, 0.98, 0.99, 1.00, 1.01, 1.03]
- random noise fraction = 0.025 with seed = 42

This construction is not intended to reproduce a licensed feeder’s exact dispatch history. Its purpose is to provide a transparent island-like load profile that preserves explicit assumptions about diurnal shape, seasonal variation, commercial demand share, and tourism-sensitive peaks.

S3. Outage-setting rationale and robustness note

The base resilience setting assumes 4 outage events per year, each with 48 h duration, a critical-load fraction of 0.55, and diesel unavailability during the outage windows. This should be interpreted as a severe island contingency stress case that represents fuel-logistics disruption rather than a universal assumption for all outages.

The base resilience results are:

- battery_only: CLSR = 1.000, EENS = 0.0 MWh, survivable outage duration = 48 h
- diesel_battery: CLSR = 0.643, EENS = 45.7 MWh, survivable outage duration = 3 h
- battery_hydrogen: CLSR = 1.000, EENS = 0.0 MWh, survivable outage duration = 48 h

Reviewer-facing robustness message: the diesel-unavailable setting is used to test whether hydrogen materially changes the resilience envelope under severe local-resource constraints. The paper’s claim is therefore conditional and screening-oriented. It does not assert that diesel is unavailable in every real outage, nor that the reported ranking would be identical under all diesel-support assumptions.

S4. Screened parameter ranges and rationale

Table S4 summarizes the exogenous dimensions used for threshold screening.

Dimension		Values used			Rationale
Delivered cost	diesel	180, 380, 700	USD/MWh		Screens low, base, and high fuel-logistics conditions for remote island supply chains.
Outage duration		12, 48, 96 h			Captures short, severe, and prolonged outage conditions relevant to resilience screening.
Hydrogen multiplier	CAPEX	0.45, 0.70, 1.00			Tests how threshold movement responds to lower hydrogen subsystem cost assumptions.
Carbon price		0, 150, 300	USD/tCO ₂		Represents a neutral case plus progressively stronger carbon-exposure stress.

At the base outage duration of 48 h and carbon price of 150 USD/tCO₂, the annual cost difference between battery–hydrogen and diesel–battery is positive at low diesel cost and high hydrogen CAPEX, but becomes negative when delivered diesel cost reaches 700 USD/MWh and the hydrogen CAPEX multiplier is reduced to 0.70 or 0.45. This is the threshold logic visualized in the main-text heat map.

S5. Reproducibility workflow and repository-ready contents

The submission package is backed by a repository-ready workflow containing:

- `configs/scenarios.json` for case, cost, resilience, and sensitivity assumptions
- `data/philippines_offgrid_8760.csv` and `data/synthetic_island_8760.csv`
- dispatch model modules in `model/`
- reproducible scripts in `scripts/`
- output tables in `results/`
- generated figures in `figures/`

The exact Windows commands documented in the repository README are:

```
.\.venv\Scripts\python scripts\generate_synthetic_data.py
.\.venv\Scripts\python scripts\run_scenarios.py
.\.venv\Scripts\python scripts\run_sensitivity.py
.\.venv\Scripts\python scripts\plot_results.py
```

This supplementary file is intended to reduce reviewer ambiguity about portfolio definitions, scenario inputs, and the reproducibility status of the paper. A public repository release can be created upon acceptance without changing the scientific content of the submitted manuscript.